Parcel100 User Manual

Offline Parcel GeoJSON Factory for ArcGIS Earth

Purpose

Parcel100 creates up to 100 small offline parcel GeoJSON files from one Master 4 1-degree box. The files are designed for direct use in ArcGIS Earth as clickable vector layers with selected parcel attributes preserved.

The program is intended to hide the repetitive GIS/server work from the operator. You provide a geographic Master 4 box and a public ArcGIS FeatureServer polygon layer; Parcel100 does the cell-by-cell querying and GeoJSON production.

Requirements

- Windows 10/11
- Python 3 available through the Windows py launcher
- Internet access while building GeoJSON files
- A public ArcGIS FeatureServer polygon-layer URL
- ArcGIS Earth for the intended viewing workflow

No QGIS installation is required for production and no third-party Python packages are required.

Start Parcel100

Keep these two files in the same folder:

```
Parcel100.py
Parcel100.bat
```

Double-click:

```
Parcel100.bat
```

The GUI opens as Parcel100 1.1.

1. Enter the Master 4 box

Enter exactly four whole-degree edges in this order:

```
west,east,south,north
```

Example:

```
82w,81w,30n,31n
```

The box must be exactly 1 degree wide and 1 degree tall.

Parcel100 subdivides the box into the established 100 Master 4 cells, each approximately 0.1 degree by 0.1 degree in geographic coordinates.

2. Enter the FeatureServer layer URL

The URL must point to a polygon layer, normally ending in:

```
/FeatureServer/0
```

or another numeric layer ID.

Florida preset:

```
https://services5.arcgis.com/GcvM6vDlR2gM4x31/arcgis/rest/services/FL_Parcels/FeatureServer/0
```

For another state or county, paste that jurisdiction's public parcel FeatureServer layer URL instead.

A web-map URL is not the same thing as the FeatureServer layer URL. Parcel100 needs the data layer itself.

3. Click Read Service

Click:

```
Read Service
```

Parcel100 reads the layer metadata and populates the field list.

The status line reports the number of fields and the service's reported maximum record count.

If the field list remains empty, verify that:

- the URL points directly to a FeatureServer layer;
- the service is public/reachable;
- the layer is a polygon layer.

4. Select fields to keep

The field list controls what parcel attributes are written into every GeoJSON feature.

For the Florida statewide layer, the current lean field set is:

```
OWN_NAME
PHY_ADDR1
PHY_CITY
PHY_ZIPCD
PARCEL_ID
```

Click Recommended to select those fields when they exist.

Other jurisdictions use different names, so inspect the aliases shown beside each field and select what is useful for your purpose.

Useful categories commonly include:

```
Owner name
Physical/site address
City
ZIP/postal code
Parcel/APN/PIN identifier
Public-land flag
```

Avoid selecting everything unless you truly need it. Fewer attributes produce smaller, cleaner field files and simpler ArcGIS Earth popups.

5. Choose the output folder

Use Browse or type a folder path.

Parcel100 creates the folder if necessary.

6. Coordinate precision

Default:

```
7
```

This controls the coordinate precision requested from the FeatureServer for the returned WGS 84 geometry.

The supported GUI range is 5 through 12.

For ordinary parcel display, the default is a good balance between geometry detail and file size.

7. Build the files

Click:

```
BUILD 100 GEOJSON FILES
```

Parcel100 processes the Master 4 cells in order.

For each cell it first asks the service only for matching object IDs. If there are no matching parcel IDs, the cell is marked empty and skipped.

A successful empty-cell result is different from a server failure. A network/service error stops the build instead of silently pretending the cell contains no data.

For populated cells, Parcel100 downloads the parcel features in service-sized batches and writes one GeoJSON file.

Output naming

Example Master 4:

```
82w,81w,30n,31n
```

Example files:

```
W082W081N031N030-004-Parcels.geojson
W082W081N031N030-005-Parcels.geojson
W082W081N031N030-006-Parcels.geojson
```

The three-digit cell number matches the established Master 4 cell addressing.

Only populated cells are written.

Completion message

At the end Parcel100 reports:

- number of GeoJSON files written;
- number of empty cells skipped;
- total parcel records written in the status line.

A field-tested Florida run produced:

```
55 GeoJSON files
45 empty cells skipped
663,277 parcel records written
```

A second Master 4 produced:

```
92 GeoJSON files
8 empty cells skipped
547,183 parcel records written
about 314 MB total
```

These are real-world examples, not guaranteed sizes or speeds.

Open the results in ArcGIS Earth

In ArcGIS Earth, add one or more .geojson files as local data.

The parcel polygons remain vector features. Click a parcel to open the feature popup and inspect the selected fields.

A practical offline stack is:

```
TPKX imagery
+ local Parcel100 GeoJSON
+ other operational vector layers/pins
```

Do not flatten the parcel layer into raster MBTiles/TPKX if you want the parcel attributes to remain clickable.

Why the files are split

GeoJSON is not a tiled database like MBTiles/TPKX. Very large GeoJSON files can take substantial time and memory to parse in a viewer.

The 100-cell design keeps each feature collection geographically small and lets the operator load only the area needed.

It also keeps FeatureServer queries manageable and naturally skips ocean/out-of-state/out-of-coverage cells.

Boundary behavior

Parcel100 queries each cell using an intersects spatial relationship.

A parcel crossing a cell edge can therefore appear in both neighboring GeoJSON files. This is intentional. Each cell remains independently useful and retains the full parcel geometry.

Other states

Parcel100 is not limited to Florida.

To try another jurisdiction:

1. Find a public ArcGIS parcel FeatureServer polygon layer.
2. Paste the layer URL into Parcel100.
3. Click Read Service.
4. Select that jurisdiction's owner/address/parcel fields.
5. Enter a Master 4 covering the desired area.
6. Build.

Coverage and schema vary widely. Some states publish statewide parcel layers; others publish county-by-county layers.

Troubleshooting

Read Service fails

Check the URL in a browser. It should resolve to an ArcGIS REST layer page and report polygon geometry.

Build stops on one cell

A timeout, service error, or connectivity problem is treated as an error. Re-run after confirming the source service is available.

ArcGIS Earth shows nothing

Confirm the GeoJSON was produced by Parcel100 and that the view is near the Master 4 area. Parcel100 requests output geometry in EPSG:4326 / WGS 84.

Popup has too much information

Rebuild with fewer selected fields.

Some parcels appear in two files

Expected at cell boundaries. Parcel100 preserves complete parcel geometry instead of clipping parcels to the cell edge.

Data caution

Parcel GIS is reference data. It can be stale, generalized, misaligned, or different from a surveyed/legal boundary. Owner/address data may represent assessment records rather than current occupancy.

For critical property decisions, verify against the responsible property appraiser, recorder, survey, deed, or other authoritative source.